UBER FARE PRICE PREDICTION

TOP 60 VIVA QUESTIONS & ANSWERS | Practical 1 · Machine Learning

How to use this revision sheet: Your code comments cover what each line does. These 60 selected questions focus on the concepts, formulas, comparisons, alternatives, applications and tricky viva points that are easy to forget.

A. Machine Learning fundamentals

1. What is Machine Learning, and what are its main types?

Machine Learning is a branch of AI that enables computers to learn patterns from data and make predictions without explicitly programming every rule. Its main types are Supervised Learning, Unsupervised Learning and Reinforcement Learning.

2. What type of ML is used in this practical? Why?

Supervised Learning, because the dataset contains input features (ride details) and labelled outputs (actual fare amounts).

3. What is Regression? How is it different from Classification?

Regression predicts continuous numerical values; Classification predicts discrete categories. Example: fare = ₹250 is regression; ride category = Expensive is classification.

4. What are features and the target variable in our practical?

Features (X): pickup longitude, pickup latitude, drop-off longitude, drop-off latitude, passenger count and distance. Target (y): fare_amount.

5. What is the difference between AI, ML and Deep Learning?

AI is the broad field of intelligent machines. ML is a subset of AI that learns from data. Deep Learning is a subset of ML using multilayered neural networks.

B. Data preprocessing and outliers

6. What is Data Preprocessing? Why is it important?

It is the process of cleaning and transforming raw data before model training. It handles missing values, duplicates, invalid records and outliers to improve data quality.

7. What is the difference between dropna() and drop_duplicates()? What are alternatives to removing missing values?

dropna() removes rows with missing values; drop_duplicates() removes repeated records. An alternative is imputation using mean, median, mode or KNN imputation.

8. What are latitude and longitude? What are their valid ranges?

Latitude measures north-south position (-90° to +90°). Longitude measures east-west position (-180° to +180°).

9. What is an outlier? Why can it affect machine learning models?

An outlier is an unusually extreme observation. It can distort statistical calculations and influence model training, especially algorithms based on squared errors.

10. What is IQR? Explain its formula and purpose.

IQR (Interquartile Range) measures the spread of the middle 50% of data. IQR = Q3 − Q1, where Q1 is the 25th percentile and Q3 is the 75th percentile.

11. What are the IQR outlier detection limits?

Lower limit = Q1 − 1.5 × IQR. Upper limit = Q3 + 1.5 × IQR. Values outside these limits are potential outliers.

12. What is the difference between IQR and Z-score?

IQR is based on quartiles and is more robust to extreme values; Z-score is based on mean and standard deviation and is more sensitive to outliers. $Z = (x - \mu) / \sigma$. A common threshold is $|Z| > 3$.

13. What are other methods of detecting outliers?

Boxplot, Z-score, Isolation Forest, Local Outlier Factor (LOF) and DBSCAN.

14. Should every outlier be removed?

No. Some outliers are genuine observations. Investigate them before removal because removing valid data can reduce performance on unusual but real cases.

C. Haversine formula and correlation

15. What is the Haversine formula? Why is it used?

It calculates the great-circle distance between two geographical coordinates on a spherical Earth. Here it calculates distance between Uber pickup and drop-off locations.

16. Write the Haversine formula and explain its variables.

$a = \sin^2(\Delta\varphi/2) + \cos(\varphi_1)\cos(\varphi_2)\sin^2(\Delta\lambda/2)$; $c = 2 \arcsin(\sqrt{a})$; $d = R \times c$. Here d is distance, R is Earth's radius (6371 km), φ is latitude in radians, λ is longitude in radians, and $\Delta\varphi$ and $\Delta\lambda$ are their differences.

17. Why do we convert degrees into radians?

NumPy trigonometric functions expect angles in radians. Conversion: radians = degrees $\times \pi / 180$.

18. What is the difference between straight-line distance and road distance?

Straight-line distance is the shortest geographical distance between two points; road distance is the actual distance travelled along roads. Haversine calculates straight-line distance.

19. What are alternatives to Haversine, and where is it used?

Alternatives include Vincenty's formula, geodesic calculations and road-routing APIs. Applications include GPS navigation, ride-hailing, aviation, fleet management and delivery services.

20. What is correlation? Explain positive, negative and zero correlation.

Correlation measures the strength and direction of association between two variables. +1 is perfect positive linear correlation, −1 perfect negative linear correlation, and 0 means no linear correlation.

21. What is Pearson correlation? What are its alternatives?

Pearson measures linear association. Spearman measures monotonic association using ranks; Kendall's Tau measures rank association using concordant and discordant pairs.

22. What is a correlation matrix and why use a heatmap?

A correlation matrix shows coefficients between pairs of numerical variables. A heatmap uses colours to make strong positive and negative relationships easy to identify.

23. Why use a heatmap instead of a bar graph? What are alternatives?

A heatmap is suitable for showing relationships between many pairs of variables. A bar graph compares quantities across categories. Alternatives include scatter plots, pair plots and correlation tables.

24. What is the difference between correlation and causation?

Correlation means two variables are associated. Causation means one variable directly causes a change in another. Correlation does not prove causation.

25. What is multicollinearity? How can it be detected?

Multicollinearity occurs when independent variables are highly correlated with each other. It can be detected using correlation matrices and Variance Inflation Factor (VIF). $VIF = 1 / (1 - R^2)$.

D. Train-test split and feature engineering

26. What is the difference between training data and testing data?

Training data teaches the model. Testing data evaluates how well it predicts unseen observations.

27. Why do we use an 80:20 split?

80% is used for training and 20% for testing. It provides a practical balance between learning and evaluation.

28. What is random_state=42? Why is it used?

It fixes the random seed so the same split or random operations can be reproduced. 42 is arbitrary, not a special mathematical constant.

29. What is feature engineering? Is distance calculation an example?

Feature engineering creates or transforms useful input variables. Yes: distance is created using pickup and drop-off coordinates.

30. What is feature scaling? Explain normalization and standardization. Is it used here?

Feature scaling brings numerical features to comparable scales. Min-Max normalization: $(X - Xmin)/(Xmax - Xmin)$. Standardization: $(X - \mu)/\sigma$. Scaling is not performed in this practical.

31. Is feature scaling compulsory for Linear Regression and Random Forest?

No. Linear Regression can work without scaling, though scaling can improve numerical conditioning. Random Forest generally does not require scaling because it uses decision-tree splits.

32. What is data leakage? Does our practical have potential leakage?

Data leakage occurs when information from outside the proper training process influences training. Yes: fare-based IQR limits are calculated and applied before splitting. A rigorous pipeline should determine preprocessing limits using training data only.

E. Linear Regression

33. What is Linear Regression and why is it used?

It is a supervised regression algorithm that models a linear relationship between input variables and a continuous target. Here it predicts Uber fares.

34. What is the difference between Simple and Multiple Linear Regression?

Simple Linear Regression uses one independent variable. Multiple Linear Regression uses two or more independent variables.

35. Write the equations of Linear Regression.

Simple: $y = mx + c$. Multiple: $y = b_0 + b_1x_1 + \ldots + b_nx_n$. Here y is predicted output, b_0 is intercept, b values are coefficients and x values are features.

36. What are slope, intercept and regression coefficients?

Slope is the change in predicted output for a unit change in input in simple regression. Intercept is the predicted output when inputs are zero. Coefficients estimate each feature's effect while holding other features constant.

37. How does Linear Regression find the best-fitting line?

It commonly uses Ordinary Least Squares (OLS), minimizing the sum of squared residuals: $SSE = \Sigma(y_i - \hat{y}_i)^2$.

38. What are the assumptions of Linear Regression?

Linearity, independence of errors, constant error variance (homoscedasticity), approximately normal errors for conventional statistical inference, and no severe multicollinearity.

39. What are the advantages and disadvantages of Linear Regression?

Advantages: simple, interpretable, fast and computationally efficient. Disadvantages: assumes linear relationships, is sensitive to outliers and may underfit complex nonlinear data.

40. What is the difference between parametric and non-parametric algorithms?

Parametric algorithms assume a model form and learn a fixed set of parameters; Linear Regression is parametric. Non-parametric algorithms make fewer assumptions about functional form; Random Forest is generally considered non-parametric.

F. Random Forest Regression

41. What is Random Forest Regression?

A supervised ensemble algorithm that combines multiple decision trees and averages their predictions to predict continuous numerical values.

42. What is a Decision Tree? Explain its components.

A tree-like model that makes predictions using conditions. Components: root node (start), decision nodes (tests), branches (outcomes) and leaf nodes (final predictions).

43. How does Random Forest work?

It creates multiple trees, trains them on randomized bootstrap samples, considers random feature subsets at splits, obtains tree predictions and averages them for regression.

44. What is Ensemble Learning? What is Bagging?

Ensemble Learning combines multiple models to improve predictions. Bagging (Bootstrap Aggregating) trains models on bootstrap samples and combines their predictions. Random Forest uses bagging.

45. Why is it called Random Forest?

Because it creates a collection of decision trees using randomness in data sampling and feature selection.

46. What does n_estimators=50 mean? How is the final prediction calculated?

It means the model contains 50 decision trees. Final regression prediction = (sum of all tree predictions) / number of trees.

47. What is the difference between Bagging and Boosting?

Bagging trains models independently and commonly reduces variance; Random Forest is an example. Boosting trains models sequentially to improve earlier errors; XGBoost and AdaBoost are examples.

48. What are the advantages and disadvantages of Random Forest?

Advantages: captures nonlinear relationships, handles complex patterns, is usually more stable than one tree and needs little preprocessing. Disadvantages: more computation and memory, less interpretability, and it can still overfit.

49. What are alternatives to Random Forest Regression?

Gradient Boosting, XGBoost, Decision Tree Regression, Support Vector Regression (SVR) and KNN Regression.

50. What is feature importance in Random Forest?

It estimates how much each feature contributes to predictions according to an importance measure. In Scikit-learn, access it using rf_model.feature_importances_.

G. Model evaluation and advanced concepts

51. Explain MAE with its formula.

Mean Absolute Error is the average absolute difference between actual and predicted values: $MAE = (1/n)\ \Sigma|y_i - \hat{y}_i|$. Lower is better.

52. Explain MSE and why errors are squared.

Mean Squared Error is the average squared difference: $MSE = (1/n)\ \Sigma(y_i - \hat{y}_i)^2$. Squaring prevents positive and negative errors cancelling and penalizes large errors more.

53. Explain RMSE and its formula.

Root Mean Squared Error is the square root of MSE: RMSE = √MSE. It expresses error in the target's original unit.

54. Explain R^2 score and its formula.

R^2 (Coefficient of Determination) measures variation in the target explained by the model: $R^2 = 1 - SSres/SStot$. A score of 1 is perfect; 0 matches the mean baseline; negative is worse than that baseline.

55. What is the difference between MAE, MSE and RMSE?

MAE averages absolute errors; MSE averages squared errors; RMSE is the square root of MSE. MAE and RMSE use the target's original unit; MSE uses squared units. All should generally be lower.

56. Can R^2 be negative? Can RMSE be smaller than MAE?

R^2 can be negative if the model performs worse than predicting the mean. RMSE cannot be smaller than MAE for the same observations.

57. What are overfitting and underfitting? How can they be prevented?

Overfitting learns training data too closely and performs poorly on unseen data. Underfitting is too simple to learn important patterns. Use regularization, cross-validation, suitable model complexity, tuning and representative data.

58. What is Cross-Validation? What is K-Fold Cross-Validation?

Cross-validation evaluates a model using multiple training and validation splits. K-Fold divides data into K parts and trains K times, using a different part for validation each time.

59. What is the difference between model parameters and hyperparameters? Give examples.

Parameters are learned during training, such as Linear Regression coefficients. Hyperparameters are set before training, such as n_estimators, random_state, max_depth and min_samples_split.

60. Which model is better for Uber fare prediction, and how can this practical be improved?

There is no universally better model. Choose based on test MAE, RMSE and R^2, plus computation and interpretability. Improvements: add traffic, surge pricing and booking time; use road distance; try XGBoost; tune with cross-validation; and prevent data leakage.

FINAL FORMULA REVISION SHEET

The key equations to recognize and explain in the viva

1. IQR	**$IQR = Q3 - Q1$**
2. Outlier limits	**$Lower = Q1 - 1.5 \times IQR$** **$Upper = Q3 + 1.5 \times IQR$**
3. Z-score	**$Z = (x - \mu) / \sigma$**
4. Degrees to radians	**$Radians = Degrees \times \pi / 180$**
5. Haversine	**$a = \sin^2(\Delta\varphi/2) + \cos(\varphi_1)\cos(\varphi_2)\sin^2(\Delta\lambda/2)$** **$c = 2 \arcsin(\sqrt{a})$ | $d = R \times c$**
6. Linear Regression	**Simple: $y = mx + c$** **Multiple: $y = b_0 + b_1x_1 + \ldots + b_nx_n$**
7. OLS / SSE	**$SSE = \Sigma(y_i - \hat{y}_i)^2$**
8. MAE	**$MAE = (1/n)\ \Sigma\lvert y_i - \hat{y}_i\rvert$**
9. MSE	**$MSE = (1/n)\ \Sigma(y_i - \hat{y}_i)^2$**
10. RMSE	**$RMSE = \sqrt{MSE}$**
11. R^2	**$R^2 = 1 - SSres / SStot$**
12. Random Forest regression	**Final prediction = (Σ tree predictions) / number of trees**

QUICK MODEL COMPARISON

Linear Regression	Random Forest Regression
Fits a linear relationship using coefficients.	Combines multiple decision trees.
Simple, fast and interpretable.	Captures complex nonlinear patterns.
Can underfit nonlinear relationships.	Usually needs more computation and is less interpretable.

Metric reminder: Lower MAE, MSE and RMSE are generally better. Higher R^2 is generally better. Compare models on the same unseen test set.